

Oxford Cambridge and RSA Examinations

A Level Computer Science

H446/01 Computer Systems

Practice Paper

Time allowed: 1 hour

INSTRUCTIONS TO CANDIDATES	INFORMATION FOR CANDIDATES
• Write your name, centre number and candidate number in the boxes above. • Use black ink. • Answer all the questions. • Write your answer to each question in the space provided. • Additional paper may be used if required.	• The marks for each question are shown in brackets []. • The total number of marks for this paper is 40. • Quality of extended responses will be assessed. • This document consists of 8 pages.

For Examiner's Use
Question 1: ____/8
Question 2: ____/10
Question 3: ____/12
Question 4: ____/10
TOTAL: ____/40

Candidate Details	
Name: _____	
Centre Number: _____	Candidate Number: _____
Date: _____	

Grade Boundaries
A*: 36-40 marks A: 32-35 marks B: 28-31 marks C: 24-27 marks D: 20-23 marks

Question 1

1. Define the term 'clock speed' as used in processor performance. [2 marks]

2. Explain how increasing the number of processor cores can improve system performance. [3 marks]

3. Briefly explain what is meant by 'pipelining' in processor design and how it improves performance. [3 marks]

Question 2

1. A modern CPU has a CISC architecture.
 - (a) State what CISC stands for. [1 mark]

 - (b) Describe two characteristics of a CISC processor. [2 marks]

 - (c) Compare CISC with RISC architecture, highlighting two differences. [4 marks]

2. Explain what is meant by the Fetch-Decode-Execute cycle. [3 marks]

Question 3

1. A computer system uses a Von Neumann architecture.
 - (a) Identify two key components of the Von Neumann architecture. [2 marks]

 - (b) Explain what is meant by the 'Von Neumann bottleneck' and how it affects system performance. [3 marks]

2. Describe the purpose and function of the following registers in a processor: [4 marks]
 - (a) Program Counter (PC)

 - (b) Memory Address Register (MAR)

3. A processor has a word length of 64 bits.
 - (a) Explain what is meant by 'word length'. [1 mark]

 - (b) Describe two advantages of a larger word length in a processor. [2 marks]

Question 4

1. Describe the differences between volatile and non-volatile storage, giving one example of each. [4 marks]

2. Explain how Solid State Drives (SSDs) differ from Hard Disk Drives (HDDs) in terms of: [4 marks]
 - (a) Physical operation

 - (b) Performance characteristics

3. A user needs to choose an input device for a specific application.
 - (a) Describe one situation where a touchscreen would be more appropriate than a keyboard and mouse. [1 mark]

 - (b) Explain one advantage and one disadvantage of using voice recognition as an input method. [1 mark]